Sum and Product of Algebraic Numbers

Given α (respectively β) that satisfies the monic polynomial f (respectively g) in $R[X]$. We want to provide a construction for the polynomial satisfied by $\alpha + \beta$ and $\alpha \cdot \beta$. The construction below assumes that 2 is cancellative in R. (For example, this is true for $R = \mathbb{Z}$.)

$\alpha + \beta$

Consider the polynomials $f(X)$ and $g(T - X)$ as polynomials in the variable X with coefficients in $R[T]$. When $T = \alpha + \beta$, they have a common root $X = \alpha$. Thus, we see that $\alpha + \beta$ is a root of the resultant $h(T)$ of $f(X)$ and $g(T - X)$ when considered as polynomials in the variable X.

We only need to check that h is a monic polynomial in the variable T.

$\alpha \cdot \beta$

We write the polynomial $f(X)$ in the form $f(X) = a(X^2) + Xb(X^2)$. Then, we see that $f(X)f(-X) = a(X^2)^2 - X^2b(X^2)^2$ is a polynomial of the form $c(X^2)$ for some polynomial $c(T)$. Note that α^2 is a root of $c(T)$.

Similarly, we write $d(T)$ such that β^2 is a root of it.

Using the previous calculation combined with the above, we find $e(T)$ such that $(\alpha + \beta)^2$ is a root of e.

Now applying the above method we can find a polynomial k for which $(\alpha+\beta)^2 - \alpha^2 - \beta^2$ is a root. This gives the polynomial satisfied by $2\alpha\beta$.